

JEE Competency Diagnosis and Adaptive Learning Assistant

Project Title

JEE Competency Mapper: Topic-Level Diagnosis, Adaptive Testing, and AI Learning Support for Class 12 Students

Project Objective

Build a prototype system that tests a Class 12 JEE student, identifies their competency level across chapters and topics, visualizes strengths and weaknesses using bar graphs, recommends learning resources, and provides an AI chatbot that explains concepts simply and detects the student's core learning difficulty.

1. Problem Statement

Many JEE students do not know exactly **where they are weak**. They may say "I am weak in Physics" or "I am bad at Calculus," but the real issue may be more specific:

- weak in Newton's Laws application questions
- poor understanding of integration by substitution
- confusion between mole concept and stoichiometry
- unable to apply formulas in multi-step problems
- making calculation mistakes
- unable to solve higher-order Bloom-level questions

This project will create a system that diagnoses student competency at three levels:

1. **Subject level:** Physics, Chemistry, Mathematics
2. **Chapter level:** e.g., Electrostatics, Organic Chemistry, Calculus
3. **Topic level:** e.g., Coulomb's Law, SN1/SN2, Definite Integration

The system will then recommend suitable study materials and provide chatbot-based support.

2. Team Size and Role Split: 3 Interns

Intern	Main Role	Responsibilities
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Intern 1	Question Bank & Competency Model	Create question-bank structure, Bloom/hardness tagging, scoring logic, topic mapping
Intern 2	Adaptive Test Engine & Dashboard	Build test flow, student scoring, competency calculation, bar graphs, progress dashboard
Intern 3	Learning Recommendation & Chatbot	Map weak topics to YouTube/PDF/notes, build concept-explanation chatbot, tutor escalation logic

All three students should jointly work on testing, demo, final report, and sample dataset.

3. Core Idea

The system should start with a **diagnostic test**. Based on performance, it should go deeper into weak chapters and topics.

Example Flow

Student logs in

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Takes overall diagnostic test

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System identifies weak chapters

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Student gets chapter-level test

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System identifies weak topics

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Student gets topic-level competency score

↓

Dashboard shows bar graphs

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System recommends videos, PDFs, notes, practice questions

↓

Chatbot explains concepts and asks follow-up questions

↓

If student is still confused, system suggests live tutor support

4. Question Bank Design

Each question in the question bank should have metadata.

Sample Question Bank Format

```
{  
  "question_id": "PHY_MECH_NLM_001",  
  "subject": "Physics",  
  "chapter": "Mechanics",  
  "topic": "Newton's Laws of Motion",  
  "subtopic": "Free Body Diagram",  
  "question_text": "A block of mass 5 kg is placed on a rough inclined plane...",  
  "options": ["A", "B", "C", "D"],  
  "correct_answer": "B",  
  "solution": "The forces acting on the block are...",  
  "bloom_level": "Apply",  
  "difficulty": "Medium",  
  "marks": 4,  
  "estimated_time_sec": 120,  
  "concept_tags": ["force", "friction", "inclined plane"],  
  "common_errors": [  
    "Missing friction direction",  
    "Resolving weight incorrectly",  
    "Using sin and cos incorrectly"
```

]

}

5. Bloom Level and Hardness Tagging

Questions should be tagged using two dimensions:

A. Bloom Level

Bloom Level	Meaning in JEE Context
Remember	Recall formula, definition, fact
Understand	Explain concept, identify relation
Apply	Use formula in standard problem
Analyze	Multi-step problem, compare methods, break into parts
Evaluate	Decide best method, identify incorrect reasoning, judge assumptions

“Create” is not needed for JEE diagnostic testing.

B. Hardness Level

Hardness	Meaning
Easy	Direct formula or basic concept
Medium	Standard JEE-level application
Hard	Multi-step, mixed concepts, high reasoning
Very Hard	Advanced JEE / Olympiad-like

6. Competency Calculation

Each student should get a competency score from **0 to 100** for each topic.

Suggested Formula

Topic Competency Score =

$0.45 \times \text{Accuracy Score}$

$+ 0.20 \times \text{Difficulty Score}$

$+ 0.20 \times \text{Bloom Score}$

$+ 0.10 \times \text{Speed Score}$

$+ 0.05 \times \text{Consistency Score}$

Explanation

Component	Meaning
Accuracy Score	How many questions were answered correctly
Difficulty Score	Whether the student can solve medium/hard questions
Bloom Score	Whether the student can answer higher-order questions
Speed Score	Whether the student solves within expected time
Consistency Score	Whether performance is stable across attempts

7. Competency Levels

Score Range	Competency Level	Interpretation
0–30	Beginner	Student lacks basic understanding
31–50	Developing	Student knows some ideas but cannot apply reliably
51–70	Intermediate	Student can solve standard problems but struggles with hard ones

71–85	Proficient	Student is JEE-ready for this topic
86–100	Advanced	Student can handle difficult and multi-concept problems

8. Dashboard and Bar Graphs

The dashboard should show competency visually.

A. Subject-Level View



B. Chapter-Level View

Example: Physics

Chapter	Competency
Mechanics	68
Electrostatics	42
Optics	75
Thermodynamics	58
Modern Physics	81

C. Topic-Level View

Example: Electrostatics

Topic	Competency
Coulomb's Law	72

Electric Field	48
Electric Potential	39
Gauss Law	31
Capacitors	55

D. Weakness Diagnosis

The system should generate statements like:

Your main weakness in Electrostatics is not formula memory.

You remember Coulomb's Law, but you struggle to apply it in multi-charge systems.

You also need more practice in Electric Potential and Gauss Law.

9. Learning Resource Recommendation

For each weak topic, the system should recommend:

1. **YouTube video**
2. **Short PDF note**
3. **Solved examples**
4. **Practice questions**
5. **Simple explanation from chatbot**

Resource Metadata Format

```
{
  "resource_id": "PHY_ELECTRIC_FIELD_VIDEO_001",
  "subject": "Physics",
  "chapter": "Electrostatics",
  "topic": "Electric Field",
  "resource_type": "YouTube Video",
  "title": "Electric Field Basics for JEE",
}
```

```
"url": "https://youtube.com/...",  
"level": "Beginner",  
"language": "English/Hindi",  
"duration_min": 18,  
"recommended_for": ["Beginner", "Developing"],  
"concept_tags": ["electric field", "point charge", "superposition"]  
}
```

Recommendation Logic

Student Level	Recommended Material
Beginner	Simple video + basic notes + solved examples
Developing	Concept video + formula sheet + easy practice
Intermediate	JEE-level examples + mixed practice
Proficient	Advanced problems + timed practice
Advanced	Previous year questions + hard challenge sets

10. Chatbot Functionality

The chatbot should not only answer questions. It should **diagnose why the student is stuck**.

Chatbot Capabilities

1. Explain concepts in simple language.
2. Ask small follow-up questions.
3. Identify misconception.
4. Give step-by-step hints.
5. Recommend resources.
6. Suggest live tutor support when needed.

Example Chatbot Interaction

Student: I don't understand Gauss Law.

Chatbot: No problem. First tell me: do you understand what electric flux means?

Student: Not really.

Chatbot: That may be the core issue. Gauss Law becomes easier once flux is clear. Imagine electric field lines passing through a surface. Flux measures how much field passes through that surface. Let us start with that before solving Gauss Law problems.

Core Issue Detection

The chatbot should classify the student's issue into one of these categories:

Issue Type	Example
Concept gap	Student does not understand electric flux
Formula confusion	Student remembers formula but uses wrong one
Application weakness	Student knows concept but cannot solve problems
Multi-step reasoning issue	Student cannot combine two concepts
Calculation error	Student makes arithmetic or algebra mistakes
Time pressure	Student knows solution but is too slow
Language comprehension issue	Student cannot understand question wording
Exam strategy issue	Student attempts hard questions first

11. Live Tutor Escalation Logic

The system should suggest a live tutor when:

- Student repeatedly fails the same topic.
- Chatbot explanation does not improve quiz performance.
- Student says "I still don't understand" multiple times.
- Competency remains below 40 after learning resources.

- Student has a serious concept gap in a high-weightage chapter.

Example Message

You have tried videos, notes, and practice questions on Gauss Law, but your score is still low.

This may need live explanation from a tutor.

We recommend a 30-minute live session focused only on:

1. Electric flux
2. Symmetry in Gauss Law
3. Choosing Gaussian surfaces

13. Suggested MVP Scope

For 6 weeks, do not try to cover the full JEE syllabus.

A good MVP scope is:

Option A: Physics MVP

Chapter	Topics
Electrostatics	Coulomb's Law, Electric Field, Electric Potential, Gauss Law, Capacitors
Current Electricity	Ohm's Law, Kirchhoff Laws, Resistivity, Circuits, Wheatstone Bridge
Mechanics	Newton's Laws, Work-Energy, Momentum, Circular Motion, Friction

Option B: Mathematics MVP

Chapter	Topics
Calculus	Limits, Differentiation, Application of Derivatives, Integration
Algebra	Quadratic Equations, Sequences, Binomial Theorem

Coordinate Geometry	Straight Lines, Circles, Parabola
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Option C: Chemistry MVP

Chapter	Topics
Physical Chemistry	Mole Concept, Thermodynamics, Equilibrium
Organic Chemistry	GOC, Isomerism, Hydrocarbons
Inorganic Chemistry	Periodic Table, Chemical Bonding, Coordination Compounds

Best starting point: **Physics MVP**, because concepts, formulas, and application levels are easy to demonstrate clearly.

14. Final Output Example

Student: Aarav Sharma

Subject: Physics

Chapter	Competency	Status
Electrostatics	44	Weak
Current Electricity	62	Improving
Mechanics	78	Strong

Weak Topic Diagnosis

Main weak topic: Gauss Law

Competency score: 31/100

Problem type: Concept gap + application weakness

You can recall the formula, but you are unable to decide the correct Gaussian surface.

You also struggle with symmetry-based reasoning.

Recommended Learning Path

Step 1: Watch beginner video on electric flux.

Step 2: Read 2-page note on Gauss Law.

Step 3: Solve 5 basic symmetry questions.

Step 4: Attempt 5 JEE-level Gauss Law problems.

Step 5: Retake topic test.

Chatbot Message

Let us not start with Gauss Law formula directly.

Your answers show that the real difficulty is understanding electric flux.

First, imagine electric field lines passing through a surface.

Now answer this: if the surface is turned sideways, will more or fewer field lines pass through it?

16. Final Deliverables

1. **Working web prototype**
2. **Question bank with 100–150 tagged questions**
3. **Competency scoring engine**
4. **Student dashboard with bar graphs**
5. **Chapter/topic weakness diagnosis**
6. **Learning resource recommendation module**
7. **Simple AI chatbot**
8. **Tutor escalation recommendation**
9. **Final internship report**
10. **Demo video**